PRANAV SHIVASHANKAR

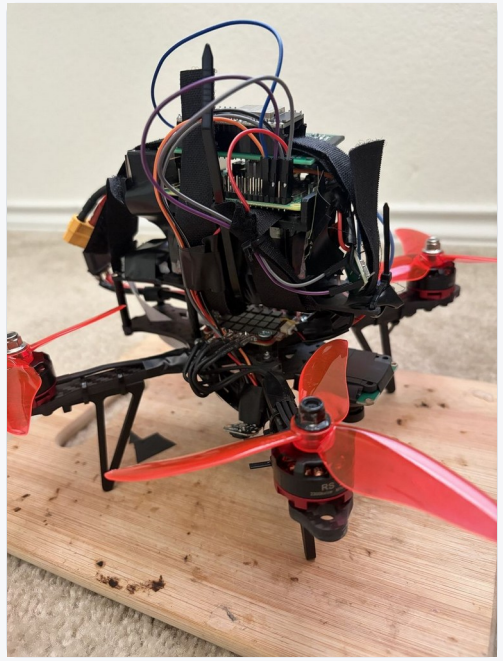
Engineering Portfolio · Summer 2027

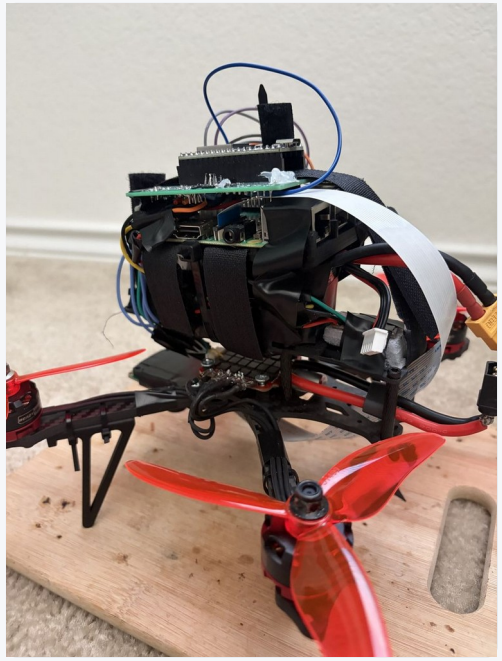
B.S. Electrical & Computer Engineering · The University of Texas at Austin · Expected May 2029

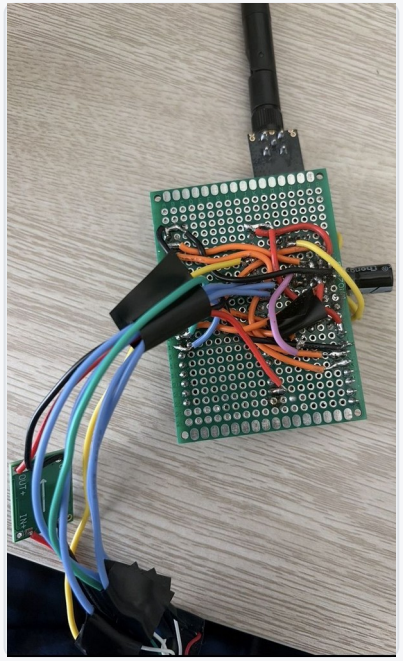
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1 Custom FPV Quadcopter: Flight Controller

Designed, built, and flight-tested from scratch · ESP32 / C++ · Jun – Aug 2026







- 500 Hz cascaded control: outer angle loop (Kp 2.45) driving an inner rate loop (Kp 0.80, Ki 0.35, Kd 0.006), bounded to 15° attitude and 200°/s yaw
- BNO085 quaternion attitude at 200 Hz over I2C, clocked at 100 kHz because the sensor stretches the clock; DShot300 to four ESCs on GPIO 32/33/25/4 via the RMT peripheral
- Anti-windup: 0.85 decay on error sign crossing, 60 ms freeze window, integration gated above 60°/s gyro rate, 240 s leak, clamped at ±60 roll and ±40 pitch
- Failsafe ladder: 350 ms grace, link loss at 700 ms, descent at 120 units/s, touchdown below 0.08 m held 250 ms, then auto-disarm

500 Hz	200 Hz	DShot300	4
CONTROL LOOP	ATTITUDE UPDATE	ESC PROTOCOL	FAILSAFE STAGES

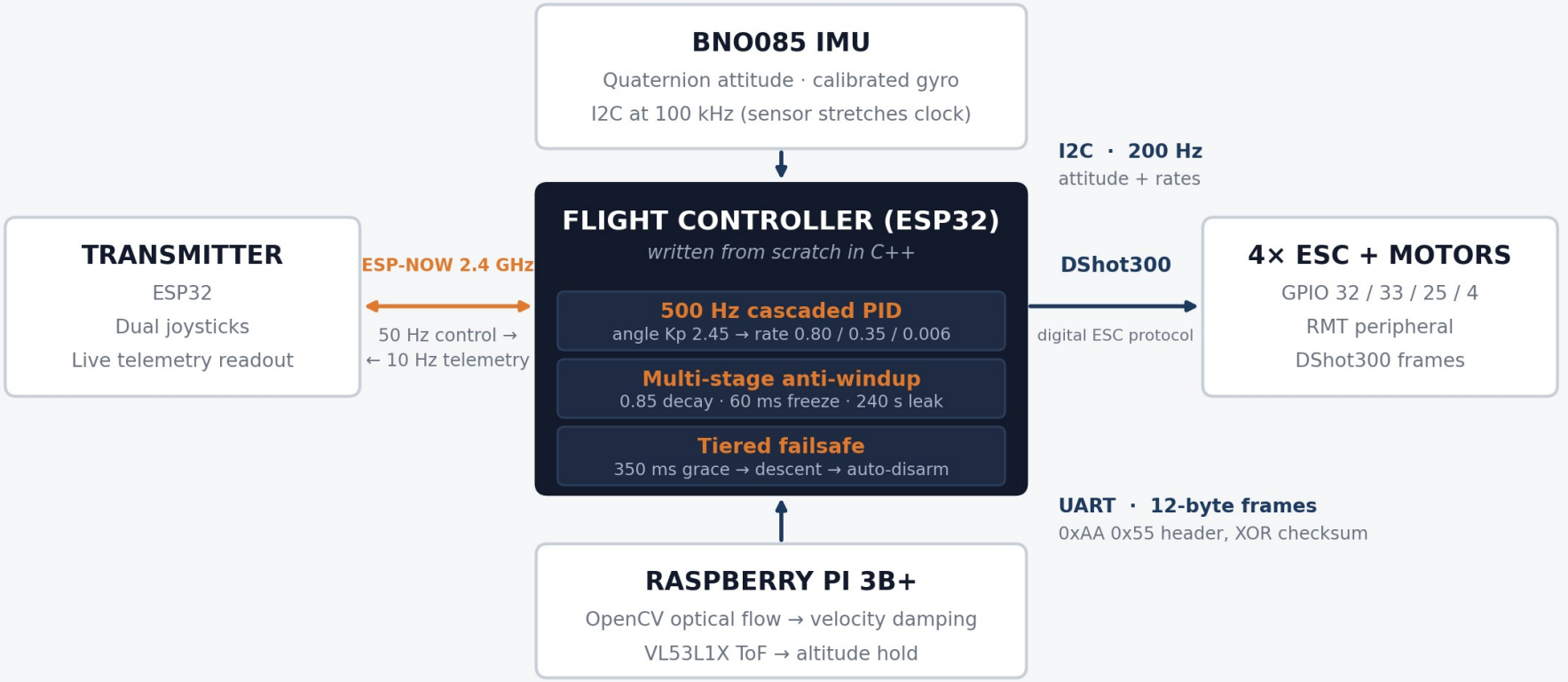
Firmware written entirely from scratch, with no flight-controller framework (Betaflight, ArduPilot, PX4).

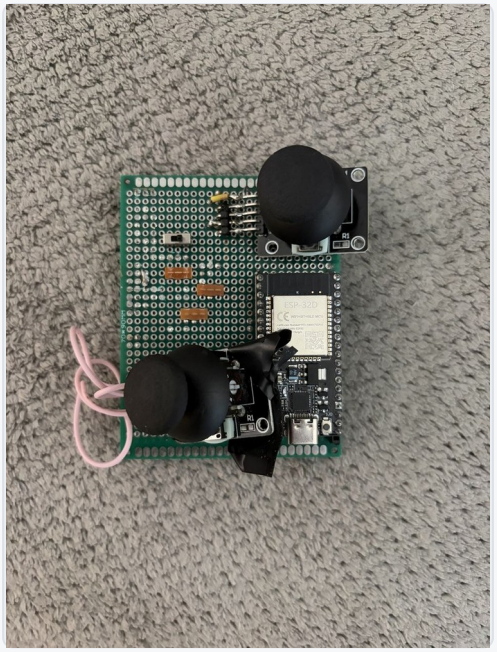
2 Quadcopter: Control and Autonomy Stack

System architecture, ground link, and companion-computer autonomy

Quadcopter System Architecture

Ground link, onboard control, and companion-computer autonomy





Dual-joystick transmitter

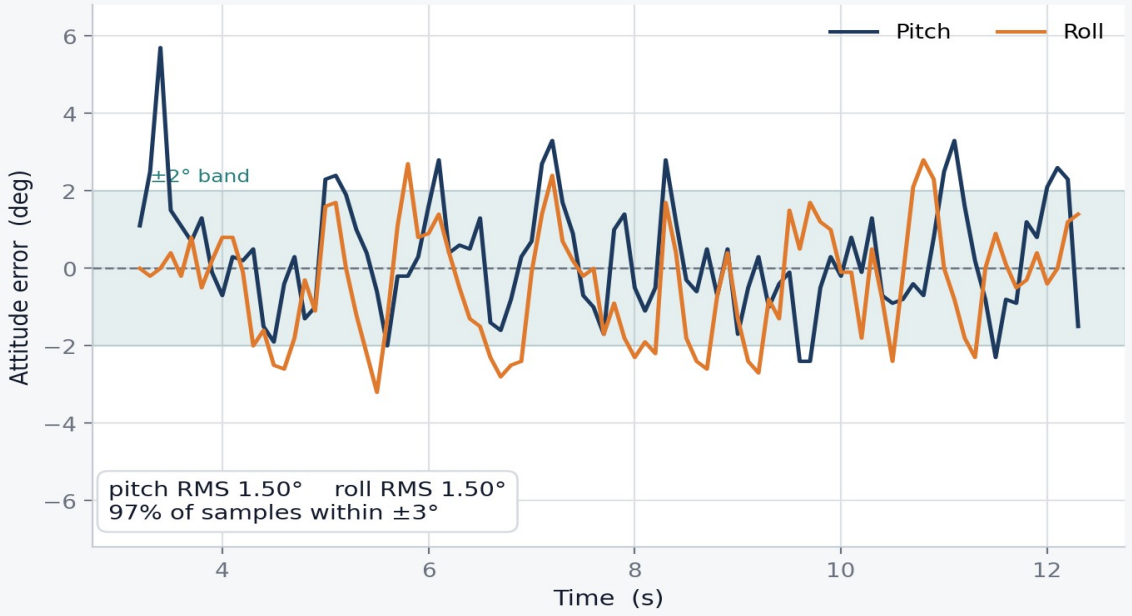
- Optical flow velocity damping: PD law at gain 0.40, engaging at flow quality 20 and dropping out at 12, authority capped at 3.5°
- VL53L1X altitude hold, PID 120 / 40 / 180, active between 0.40 m and 2.50 m with bumpless engagement
- Heading hold integrates yaw rate at Kp 1.5, limited to 60°/s and reset on stick input
- Trim solved from logged integrator steady state: PITCH_TRIM 6.0 to 8.2, ROLL_TRIM 0.95 to 0.7

3 Quadcopter: Flight Test Results

Measured from logged in-flight telemetry, not simulation

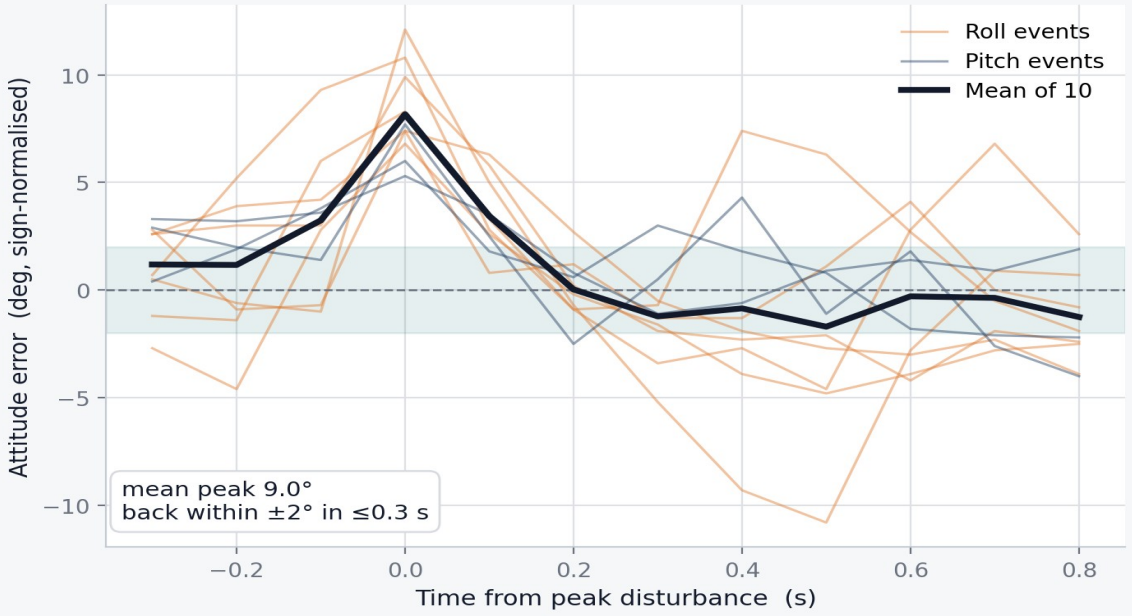
Hands-off hover: attitude hold accuracy

No pilot input on pitch or roll · 9 s of continuous hover



Disturbance recovery: 10 commanded tilts

Aircraft tilted on purpose, then released to self-level



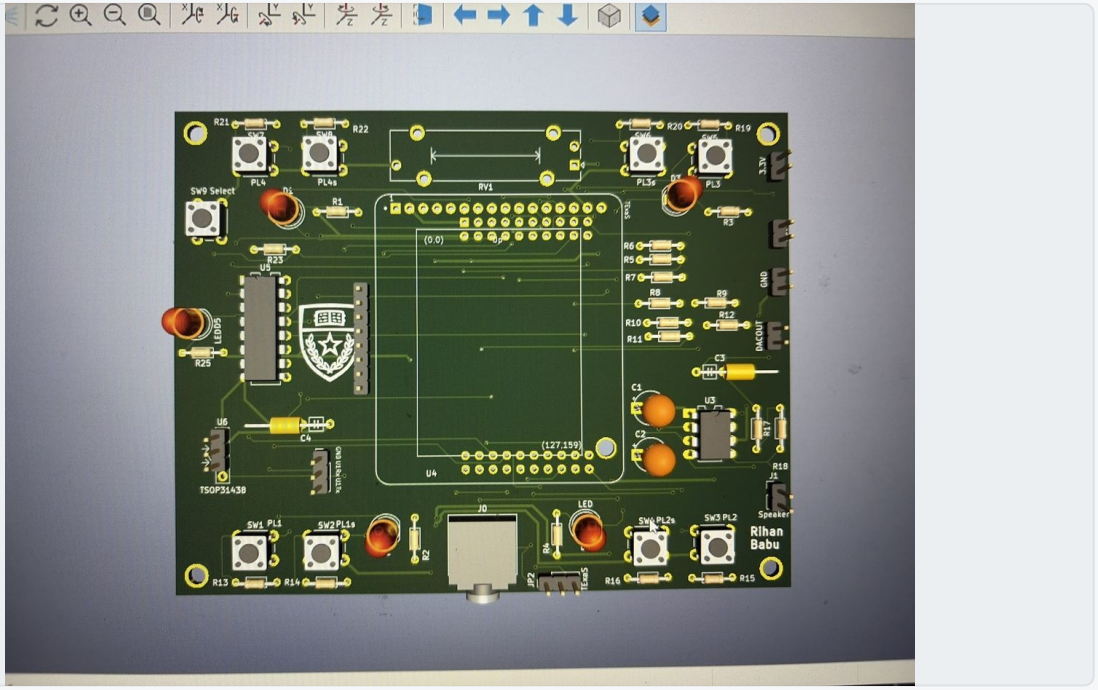
- Aircraft left to hold level with no pitch or roll input, measuring how tightly the controller keeps attitude in steady hover
- Aircraft tilted on purpose and released, measuring how quickly it drives the resulting error back to level

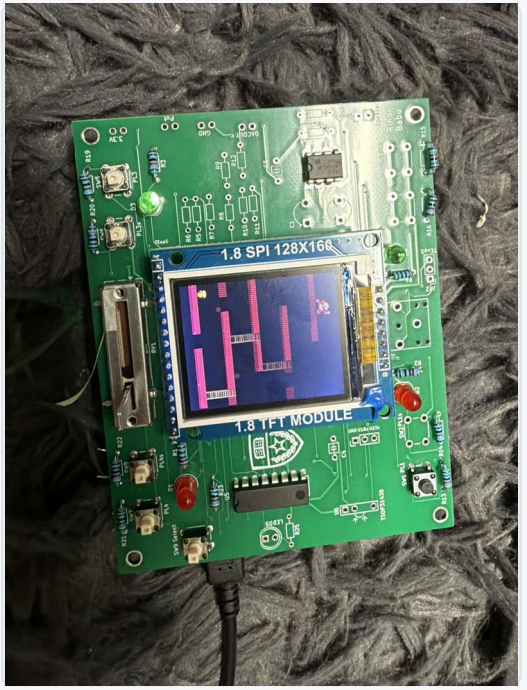
1.50°	97%	9.0°	≤0.3 s
HOVER RMS ERROR	WITHIN ±3°	MEAN DISTURBANCE	RECOVERY TO ±2°

Separate logged flights. Telemetry sampled at 10 Hz over the ESP-NOW downlink, so the 500 Hz control loop settles faster than the link can resolve.

4 Embedded Game Console: Custom PCB

TI MSPM0G3507 · ARM assembly and C · Mar – May 2026



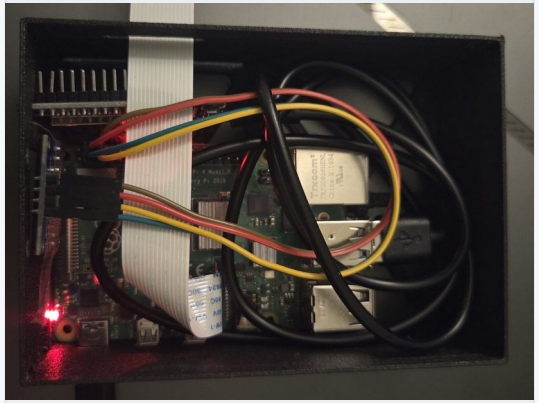


- Designed and assembled a custom PCB integrating buttons, LEDs, an LCD, a slide potentiometer, and audio output
- Wrote a Donkey Kong-style infinite platformer in ARM assembly and C
- Implemented LCD graphics rendering, a title screen, language selection, and sound effects
- Full stack from schematic capture through board bring-up to shipped firmware

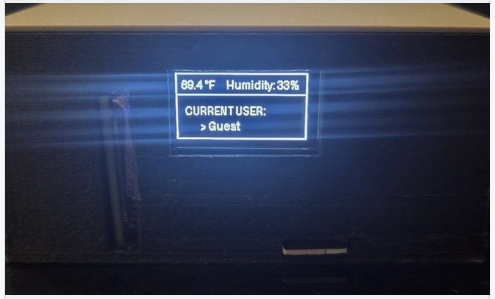
5 FridgeGuard: 3rd Place, UT IEEE Techathon

Raspberry Pi 4B + ESP32 · computer vision and BLE attribution · Apr 2026

- Shared-living-space fridge security system built and presented in a single hackathon weekend
- Captures before/after frames via 500 ms sampling between door-open and door-close events
- Groq Llama 4 Scout vision API compares image pairs against a SQLite inventory database to identify the specific item added or removed, then updates stock state
- ESP32 scans roommates' phone BLE advertisements to attribute each fridge action to a specific user



Pi 4B + wiring



OLED status display

Detection pipeline

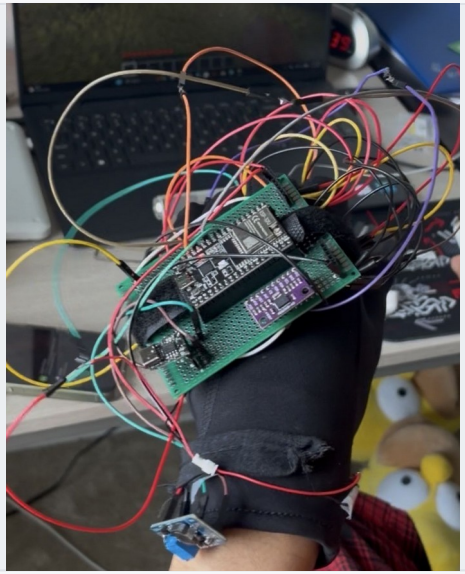
1. Door-open event detected
2. Frames sampled every 500 ms until close
3. Before/after pair sent to Llama 4 Scout
4. Diff resolved against SQLite inventory
5. BLE scan attributes action to a user

3rd	500 ms	2	48 hrs
PLACE, UT IEEE	FRAME SAMPLING	MCUs INTEGRATED	BUILD TIME

6 Gesture-Control Glove: Two Firmware Targets

ESP32 · BNO085 IMU, hall-effect and contact sensing · Feb – Apr 2026





- Four fingertip contact pads on GPIO 13/14/27/26 plus two hall-effect sensors that register a closed fist when index and ring trigger together
- BNO085 quaternion attitude at 50 Hz, zeroed against a measured neutral hand pose and guarded by a five-attempt init retry so a missing IMU disables tilt instead of hanging
- Sensor frame rotated into the control frame by summing and differencing the pitch and roll deltas, so a naturally angled hand maps cleanly to two orthogonal axes

Drone mode · ESP-NOW transmitter

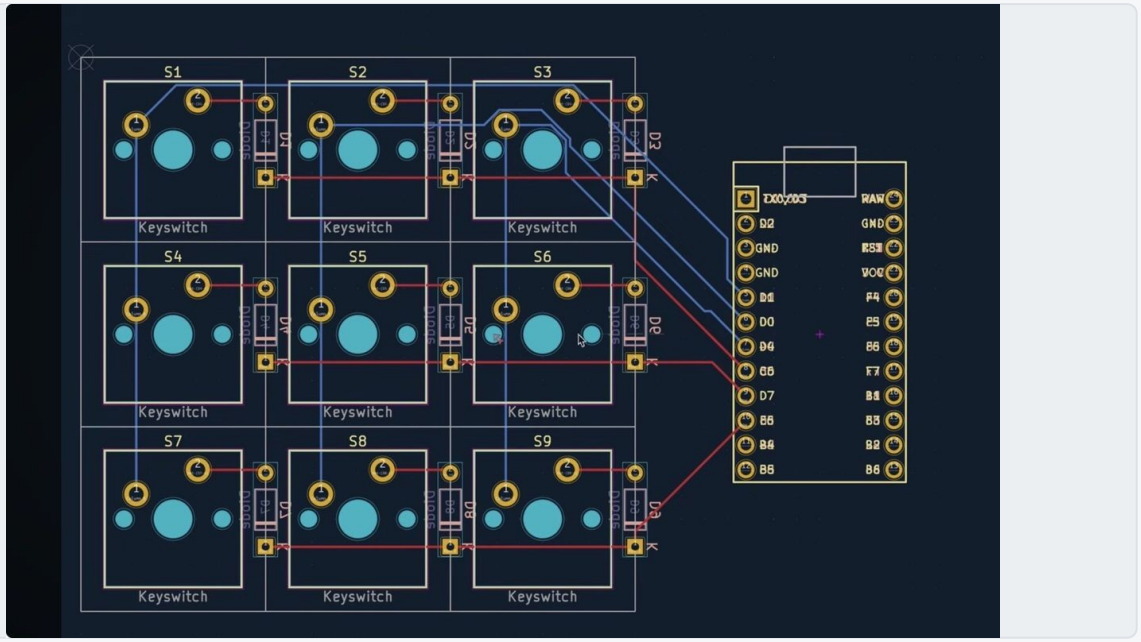
- Packed 10-byte struct byte-matched to the transmitter: pitch, roll and yaw scaled to ±1000 with a 13° deadzone and 65° full deflection, sent at 50 Hz
- Ring finger acts as a modifier that reroutes lateral tilt from roll to yaw; index commands throttle up, middle commands land
- Redundant kill on pinky or fist, while arming stays on the transmitter's physical switch by design and is never exposed to the glove

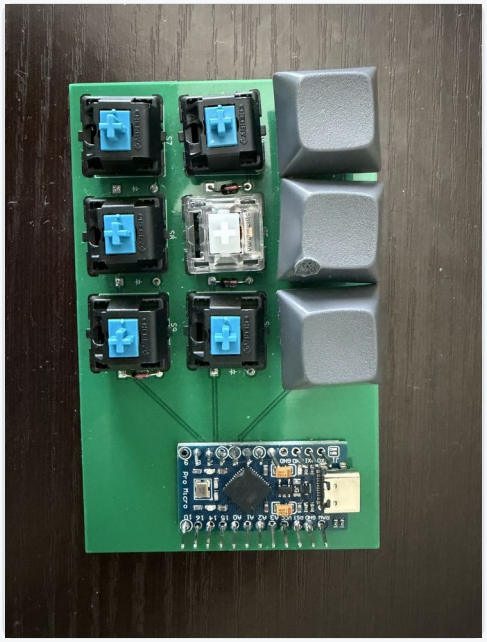
PC mode · BLE HID keyboard

- Same hardware pairs to Windows as a standard keyboard with no host-side drivers, using edge-detected press and release for true key-hold behaviour
- Tilt drives strafe and walk past a 40° threshold; index holds jump, middle latches sprint as a toggle, ring and pinky tap inventory and sneak
- Closed fist sends escape behind a one-second debounce

7 Custom Macropad: Keyboard PCB Design

KiCad · Arduino Pro Micro (ATmega32U4) · QMK firmware





- Designed a 3×3 macropad in KiCad from schematic capture through PCB layout and fabrication
- Laid out a switch matrix with per-key 1N4148 diodes to prevent ghosting and enable reliable n-key rollover
- Hand-soldered Cherry MX switches, diodes, and an Arduino Pro Micro (ATmega32U4) controller
- Flashed QMK firmware with a custom keymap, taking the board from bare PCB to a working USB HID device

Full source and design files for every project: github.com/pranavs41 · pranavshivashankar12@gmail.com